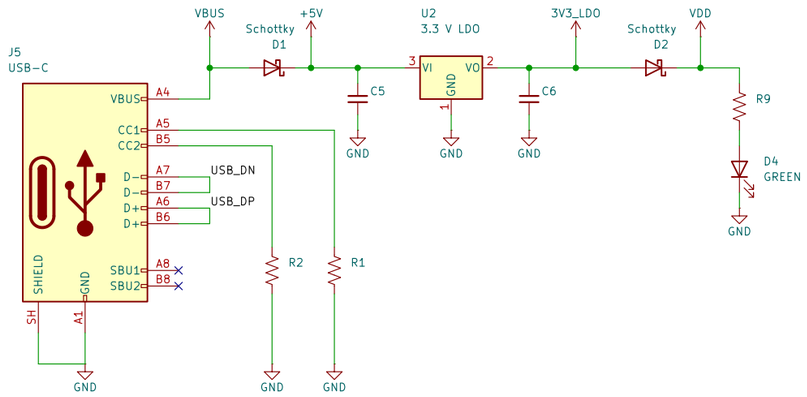
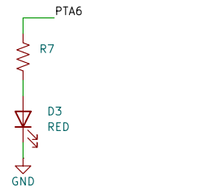
USB-C + POWER

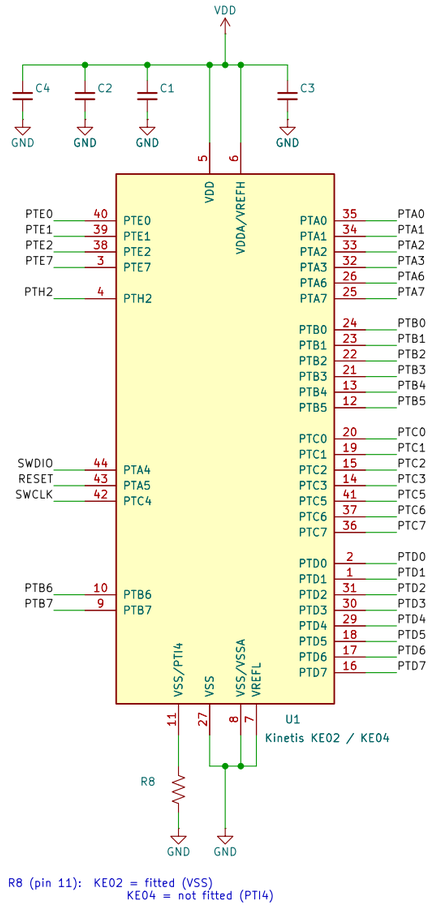


USER LED



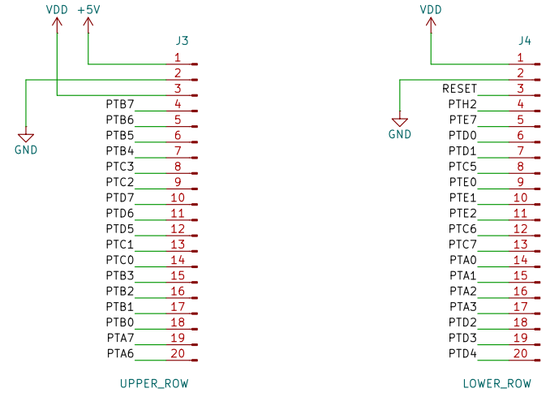
active high (pin -> R7 -> LED)

MCU (KE02 or KE04, LQFP-44)

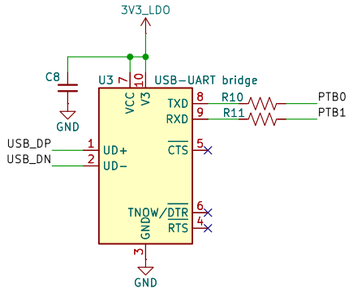


R8 (pin 11): KE02 = fitted (VSS)
KE04 = not fitted (PTI4)

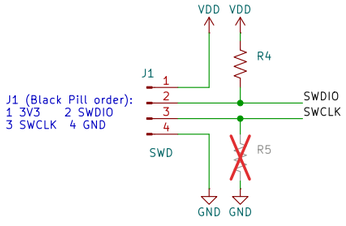
BREADBOARD HEADERS (pin 1 = USB end)



USB-SERIAL (UART0)

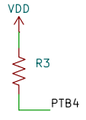


SWD + RESET

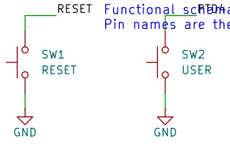


J1 (Black Pill order):
1 3V3 2 SWDIO
3 SWCLK 4 GND

NMI PULL-UP



BUTTONS



Functional schematic for reference. Component values and part numbers are omitted on purpose.
Pin names are the MCU port names; see the pinout and datasheet in the docs repository.

Title: KE02/KE04 Black Pill		Hardcode
File: KE02_KE04_Black_Pill.kicad_sch		Sheet 1 of 1 — Size: A3
Date: 2026-10-09	Rev: A	Draw by: Mauricio Mendes